



Alberta Public Health Disease Management Guidelines

Escherichia coli, Shiga Toxin-Producing (STEC) Infections



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Communicable Disease Control Branch

Public Health Division

Alberta Health

***Escherichia coli*, Shiga Toxin-Producing (STEC) Infections** | Alberta Health,
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Revisions

Revision Date	Document Section	Description of Revision
April 2025	General	- Updated name to <i>Escherichia coli</i>, Shiga Toxin-Producing Infections. - Revised entire guideline to align with current information.
	Case Definition	- Updated to align with national case definition. - Added footnotes to clarify testing requirements and guidance for interpretation of results. - Added Table 1: Test Results and Case Classification
	Reporting Requirements	Included clarification for reporting timelines of outbreaks.
	Clinical Assessment and Epidemiology	Updated section name
	Public Health Management	- Revised Table 2: Sensitive Situations or Occupations (SSO) - Added footnote for referral to the Alberta Food Regulation for specific definitions. - Revised Table 3: Case Exclusion table to include recommended specimen collection timelines. - Added Table 4: Interpretation of Microbiological Clearance Testing Results. - Revised Table 5: Contact exclusion table to only include exclusion information.
	Preventive Measures	- Consolidated prevention information. - Added links to My Health Alberta and Government of Canada Food Safety websites.

Case Definition

Includes O157 and non-O157 serotypes

(Previously *Escherichia coli* Verotoxigenic Infections)

Confirmed Case

Laboratory confirmation of infection with or without clinical illness:^(A)

- Isolation of Shiga toxin-producing *Escherichia coli* (includes but not limited to O157:H7) from an appropriate clinical specimen (e.g., stool, urine, blood).^(B)

OR

- Detection of Shiga toxin antigen or nucleic acid in an appropriate clinical specimen (dependent on the test used) ^(B) using a diagnostic test such as a Shiga toxin antigen test or a Shiga toxin nucleic acid test (NAT), such as polymerase chain reaction (PCR). ^{(C)(D)}

Probable Case

- Clinical illness^(A) in a person who is epidemiologically linked ^(E) to a confirmed case-which would include persons with haemolytic uremic syndrome (HUS)

OR

- Detection of *E. coli* O157 nucleic acid that is Shiga toxin negative or pending, with or without clinical illness, in an appropriate clinical specimen^(B) (i.e., dependent on the test used) using a NAT, such as PCR.^{(F)(G)}

^(A) Clinical illness is characterized by diarrhea (often bloody) and abdominal cramps, although asymptomatic infections do occur. Illness may be complicated by hemolytic uremic syndrome (HUS), thrombocytopenia purpura (TTP) or pulmonary edema. The microorganism may cause extra-intestinal infections.

^(B) Refer to the APL [Public Health Laboratory \(ProvLab\) Guide to Services](#) for current specimen collection and submission information.

^(C) Culture is important for clinical management and to enable molecular typing required for surveillance, outbreak detection and response. NAT negative samples can be considered culture-negative (see [Diagnosis](#)).

^(D) When reflex culture and/or Shiga toxin typing occurs, classification as a confirmed case will remain unchanged regardless of culture and typing results. A NAT/PCR-positive (NAT/PCR+) Shiga toxin and culture-negative (culture -) result for STEC is considered a confirmed case ([Table 1: Test Results and Case Classification](#)).

^(E) Epidemiologically linked includes exposure to a common source (e.g., food).

^(F) A NAT/PCR-positive (NAT/PCR+) for *E. coli* O157 and a culture-negative (culture -) result for Shiga toxin-producing *E. coli* is considered a probable case.

^(G) Note: detection of *E. coli* O157 nucleic acid is not routinely used for STEC diagnosis in Alberta at this time.

Table 1: Test Results and Case Classification

NAT/Antigen Test	Result	STEC Culture	Result	Case Classification
Shiga toxin NAT/Antigen	Positive	Completed	Positive	Confirmed
			Negative	Confirmed
		Not Completed	N/A	Confirmed
	Negative	Not Completed	N/A	Not a Case
Not Completed	N/A	Completed	Positive	Confirmed
			Negative	Not a Case
O157 NAT	Positive	Completed	Positive	Confirmed
			Negative	Probable
		Not Completed	N/A	Probable

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- MOH (or designate) of the zone, and
- Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and Indigenous Services Canada (ISC)

- The Zone MOH (or designate) where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- The Zone MOH (or designate) shall report all outbreaks to CMOH (or designate) via the [Alberta Outbreak Report Form \(AORF\)](#) as soon as possible using existing processes (e.g., CDOM or fax).
 - The initial report form must be submitted within **24 hours** of opening the outbreak investigation and the final report must be submitted within **48 hours** of closing the outbreak investigation.
- For out-of-province and out-of-country cases and/or contacts, the Zone MOH (or designate) should forward the following information to the CMOH (or designate) using existing processes (e.g., CDOM or fax) within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report (if applicable), and
 - other relevant clinical/epidemiological information.

Clinical Assessment and Epidemiology

Etiology

Escherichia coli (*E. coli*) is a gram-negative bacillus that can cause diarrheal illness.⁽¹⁾ Shiga toxin-producing *E. coli* (STEC) is a group of *E. coli* strains that cause illness by producing Shiga toxin 1 (Stx1) and/or 2 (Stx2). Strains producing Stx2 have an increased risk of bloody diarrhea and hemolytic uremic syndrome (HUS).^(2,3) STEC also expresses additional virulence factors including intimin as the principal adherence factor.^(3,4) Although O157:H7 is consistently virulent and often implicated in outbreaks, other serotypes can also cause severe illness.⁽⁵⁾

The organism has a low infective dose, ranging from 10 to 100 colony forming units (cells).⁽⁴⁾

Clinical Presentation

STEC infection ranges from asymptomatic to severe illness. Symptomatic infections generally begin with non-bloody diarrhea and severe abdominal cramps; vomiting and low-grade fever may or may not be present.^(4,5) Diarrhea may become bloody after two to three days due to onset of hemorrhagic colitis.⁽⁵⁾

The illness is often self-limited, lasting 5 to 7 days, with an average of 10 days for uncomplicated cases of STEC.^(6,7)

Complications of STEC infections include bloody diarrhea, HUS and may result in hospitalization.^(8,9) The severity of infection and risk of HUS is related to the Stx type rather than the STEC serotype.^(8,9) HUS occurs most commonly in children under 10 years old.⁽⁹⁾ HUS can develop approximately 7 – 14 days after diarrhea begins and is defined by the triad of acute renal dysfunction, thrombocytopenia and hemolytic anemia.^(3,5) Approximately 10% of patients with STEC may develop HUS and of these an estimated 30% to 60% of patients with STEC-HUS require renal replacement therapy such as dialysis with a case fatality rate between 3% to 5%.^(3,10,11)

For information, please refer to the [HUS public health disease management guideline](#).

Diagnosis

STEC infections were conventionally diagnosed by positive culture. However, the use of nucleic acid/polymerase chain reaction (NAT/PCR) tests (such as PCR for Shiga toxin) and antigen-based tests for the detection of STEC in stool is increasing, including for non-O157 strains.^(3,5,12) The diagnosis of STEC should be considered for individuals experiencing severe diarrhea (especially if bloody), HUS, thrombotic thrombocytopenic purpura (TTP) or hemorrhagic colitis.⁽⁵⁾ Early diagnosis of STEC is important for clinical management and monitoring for the development of HUS.⁽²⁾

When NAT/PCR or antigen-based tests are used for diagnosis, culture is still important for surveillance purposes and occasionally for clinical management. Culture is particularly important when the Shiga toxin type cannot be determined through NAT/PCR.⁽¹²⁾ According to [Canadian Public Health Laboratory Network \(CPHLN\)](#) guidance, culture performed on NAT-positive specimens enables confirmation of the identified bacteria and further strain characterization, including serotyping and molecular typing (e.g., whole genome sequencing) for surveillance, outbreak detection and response.^(12,13) For more information on STEC testing pathways in Alberta, please refer to [Alberta Precision Laboratories](#).

Treatment

Treatment is mainly supportive, focusing on fluid and electrolyte replacement.^(3,5) Antibiotics are generally not recommended as their use has the potential to increase Shiga toxin production/release and increase the risk of developing HUS.^(5,14,15) Antimotility agents should also be avoided, particularly in cases with bloody diarrhea.^(5,16)

Reservoir

Cattle are the principal reservoir of *E. coli*, including STEC O157:H7 and non-O157, however sheep, goats and deer may also carry STEC.⁽³⁾ Humans may serve as a reservoir for person-to-person transmission.⁽³⁾

Transmission

The predominant mode of transmission is through ingestion of contaminated food, often due to inadequate cooking (such as undercooked beef or pork) or cross-contamination during food preparation. Foods, particularly fruits and vegetables contaminated by cow manure, as well as unpasteurized milk, cheese and juice, have been documented modes of transmission.^(5,17) Waterborne transmission can occur after swimming in or drinking water contaminated with human or animal feces.^(3,5) Infection can also be acquired through contact with a carrier animal and their environment.⁽⁵⁾ Direct person-to-person transmission can also occur, and has been observed in families, daycares, and institutions.^(3,5)

Incubation Period

The incubation period for most STEC serotypes is typically 3 - 4 days with a range of 2 - 10 days.⁽³⁾ As the incubation period for individual cases can vary, clinical discretion is required when determining where and how the disease was likely acquired.

Period of Communicability

The communicable period lasts as long as the organism is being excreted. In adults, this period of communicability is generally up to seven days and in about one- third of children can be approximately three weeks.⁽³⁾ Prolonged carriage in young children under six years of age has been documented, with median shedding duration of 32 days.⁽¹⁸⁾

Host Susceptibility

Universal. However, children one to four years of age are most frequently diagnosed with infection and are at higher risk of developing HUS. Older adults who develop HUS are at increased risk of death.⁽³⁾

Incidence

While STEC infections are reported globally, they are more commonly reported in industrialized and high-income regions such as North America, Europe, Japan, Australia, and the southern cone of South America.^(3,19,20)

In Canada, STEC infections became nationally notifiable in 1991. For current case counts in Canada, refer to [the Notifiable Diseases Online](#) website.

In Alberta, *E. coli* O157:H7 became notifiable in 1983 and STEC infections became notifiable in 2010. Refer to the [Interactive Health Data Application \(IHDA\)](#) for more information.

Public Health Management

Key Investigation

- Confirm the individual meets case definition.
- Obtain a history of illness including the date of onset.
- Determine [incubation period](#) and [period of communicability](#).
- Determine if the case is in a [sensitive situation or occupation \(SSO\)](#) that poses a higher risk of transmission to others.
- Determine the possible source of infection taking into consideration the [incubation period](#), reservoir, and mode of transmission. Assessment may include:
 - a detailed food history including recent consumption of undercooked meat or other potential sources (e.g., unpasteurized milk/dairy products, unpasteurized juices, raw fruits and vegetables),
 - contact with sewage contaminated recreational water (cattle nearby),
 - consumption of untreated surface water,
 - possible exposure to farm animals (including petting zoos) or pets (including recent illness in pets or new puppy, kitten, etc. into the household),
 - exposure to fowl, domestic or wild animals,
 - exposure to raw pet treats/foods or kibble⁽²¹⁾,
 - occupational exposure (e.g., animal or meat handling),
 - high risk sexual practices, especially contact with feces,
 - recent travel or recent immigration, and/or
 - residing in areas with poor sanitation, including improper water treatment and sewage disposal either in Canada, or abroad.
- Obtain implicated food samples, if possible.
- Suspected contaminated food may be held or destroyed if not needed as part of the investigation, to prevent consumption.
- Assess for history of similar symptoms in other members of the household.
- Identify any contacts, especially those in [SSO](#), that pose a higher risk of transmission to others. Contacts include:
 - persons living in the household,
 - children and childcare workers at a child care facility (daycare, dayhome, or other child care site),
 - any other persons that may have had direct or indirect exposure to the feces of the case, and
 - individuals exposed to the same source (if identified).

Table 2: Sensitive Situations or Occupations (SSO)

SSO	Activities
Food handler ^(H)	- Touches unwrapped food^(I) to be consumed, <u>and/or</u> - Handles equipment or utensils that touch unwrapped food to be consumed*
Health care, child care or other staff	- Has contact through serving food to patients or persons in whom an intestinal infection would have particularly serious consequences. - Provides direct patient care or is involved in the care of young children, elderly or dependent persons.
Child attending a child care facility or similar facilities	Individuals who are diapered, incontinent, or require assistance with toileting.
Any individual (older child or adult)	Individuals who are diapered, incontinent, or require assistance with toileting.

* NOTE: Generally, food handlers who do not touch food, equipment or utensils in this way are not considered to pose a transmission risk; however, circumstances for each case should be assessed on an individual basis.

^(H) Please refer to the [Alberta Food Regulation](#) for definitions of “food handler” and “handle”.

^(I) Please refer to the [Alberta Food Regulation](#) for definition of “food”.

Management of a Case

- All cases should be advised of the following:
 - Be assessed by their health care provider, especially young children,
 - only take medications for treatment as recommended by a health care provider (i.e., avoid over the counter anti-diarrheal medication and antibiotics),
 - how the disease is transmitted, appropriate personal hygiene, routine infection prevention and control practices, and contact precautions,
 - avoid food preparation for others until 48 hours after symptoms have resolved,
 - avoid public recreational water (e.g., pools, lakes, ponds) until two weeks after the symptoms resolve, and
 - when to access emergency medical services.
- Hospitalized cases should be managed in accordance with hospital infection prevention and control guidelines.
- Exclusion*: Refer to [Table 3: Case Exclusion](#).

Table 3: Case Exclusion

Cases	Category	Exclusion Criteria
Symptomatic and Asymptomatic	SSO	• The MOH shall by order exclude the case until two stool specimens taken from the infected person are reported as negative (Table 4)⁽³⁾ – Collect follow-up stool specimens at least 48 hours after normal stools resume. – Collect subsequent stool specimens at least 24h after a previous negative stool OR at least 72h after a previous positive stool. • The MOH may consider reassignment to a low-risk area as an alternative to exclusion (i.e., the case would no longer be considered to be in an SSO).
Symptomatic	Non-SSO	• No exclusion required. – However, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped.
Asymptomatic	Non-SSO	• No exclusion required.

*Note: The [Enteric Transmission Risk Assessment and Exclusion Table](#) summarizes exclusions and recommendations for microbiological clearance for cases and contacts of cases with enteric illnesses who work in SSOs.

Table 4: Interpretation of Microbiological Clearance Testing Results

NAT/PCR	Culture	Interpretation
Negative	Not done	Negative stool
Positive	Positive	Positive stool
Positive	Negative	Negative stool
Not done	Positive	Positive stool
Not done	Negative	Negative stool

⁽³⁾ For an asymptomatic case who has an initial specimen that is STEC NAT/PCR (+) and culture (-), the MOH **shall** by order exclude the case until **one** additional stool specimen taken from the infected person is reported as negative, pending the case does not develop symptoms. The stool shall be collected no less than 24 hours after the initial specimen.

Management of Contacts

- [Contacts](#) should be instructed on disease transmission, appropriate personal hygiene, routine practices, and contact precautions.
 - Emphasize measures to minimize fecal-oral transmission, including strict hand hygiene, especially after using the washroom, changing diapers, and before eating or preparing/handling foods.
- Assess contacts for potential of exposure during period of communicability for the case, as well as for signs and symptoms of illness.
 - Symptomatic contacts should be referred to their health care provider for assessment and testing if clinically indicated.
 - Asymptomatic contacts should monitor themselves for gastrointestinal symptoms, maintain good hand hygiene and food handling practices and seek medical attention if symptoms develop.
- Exclusion*: Refer to [Table 5: Contact Exclusion](#).

Table 5: Contact Exclusion

Contacts	Category	Exclusion Criteria
Symptomatic	SSO	• Same as a case (see Table 3).
Asymptomatic	SSO	• The MOH may by order exclude asymptomatic contacts until one stool specimen is reported as negative.
Symptomatic	Non-SSO	• No exclusion required. – However, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped.
Asymptomatic	Non-SSO	• No exclusion.

*Note: The [Enteric Transmission Risk Assessment and Exclusion Table](#) summarizes exclusions and recommendations for microbiological clearance for cases and contacts of cases with enteric illnesses who work in SSOs.

Preventive Measures

Educate the public about the following:

- personal hygiene, especially:
 - the sanitary disposal of items containing feces,
 - careful hand washing,
 - after defecation
 - before/after sexual contact,
 - after changing another person's diaper or assisting with toileting, and
 - before/after preparing or eating food.
 - after contact with animals, animal feces, and animal environments, especially if animals are ill with diarrhea.
- accessing and drinking safe water supplies, and testing private water supplies for bacterial contamination, if suspected,
- the risk of sexual practices that permit fecal-oral contact.
- Food safety practices:
 - proper food and equipment handling, preparation, and hygiene, with a focus on avoiding cross-contamination from raw meat products,
 - avoid food handling or preparation while symptomatic until 48 hours after diarrhea has resolved.
 - thorough hand washing before and after meal preparation,
 - thoroughly wash fresh produce prior to consumption or preparation,
 - thoroughly cooking various meats (e.g., ground beef to 71°C),⁽²²⁾
 - avoid consuming raw or unpasteurized milk or other dairy products and unpasteurized juice/cider, and
 - avoid cross contamination of food by:
 - keeping uncooked meats separate from produce, cooked foods, and ready-to-eat foods.
 - thoroughly washing cutting boards, counters, knives and other utensils with soap and water after handling uncooked foods, especially after contact with raw meat,
- For more information on general food safety tips, please refer to the [My Health Alberta: Foodborne Illness and Safe Food Handling website](#) and the [Government of Canada General Food Safety Tips website](#).

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